

MEMA

COLLEGE

SYSTEM REQUIREMENTS AND DESIGN SPECIFICATION

End-to-End Admission Module

Document purpose: Implementation-ready specification for the public applicant portal, internal admissions workspace, workflow, controls, integrations, analytics, documents, reporting and UAT.

Document control	Value
Document ID	MC-ERP-ADM-SRDS-001
Version	1.1
Status	Draft for stakeholder validation
Prepared	24 August 2026
Initial target intake	September 2026
System owner	Registrar / Admissions Office
Classification	Internal - Controlled
Branding note	Mema logo and official palette to be supplied; current wordmark and teal/gold scheme are placeholders.

Approval signatures: Product Owner _____ Registrar _____ ICT Lead
_____ Date _____

Document Control and Governance

Version	Date	Description	Owner
0.1	Aug 2026	Initial consolidated requirements	Project Team
1.0	24 Aug 2026	Complete SRDS and UAT baseline	Mema College
1.1	24 Aug 2026	Added mandatory KES 1,000 application-fee gate before submission	Mema College

Approvals

Role	Name	Decision	Date	Signature
Executive Sponsor	TBD	Pending		
Registrar / Product Owner	TBD	Pending		
ICT Lead	TBD	Pending		
Finance Lead	TBD	Pending		
Data Protection Officer	TBD	Pending		

Definitions and lifecycle vocabulary

Term	Meaning
Prospect / Lead	A person who expresses interest but has not submitted an application.
Applicant	A verified account holder with at least one application record.
Submitted application	A formally lodged, immutable version of the application awaiting checks.
Shortlisted applicant	An applicant who satisfies configured screening criteria; this is not an offer.
Admitted candidate	An applicant approved for admission and issued an offer/admission letter.
Accepted candidate	An admitted candidate who accepts the offer and satisfies acceptance conditions.
Enrolled student	A candidate converted into the Student Information System after registration controls are met.
Cohort / Intake	A configured admission period such as September 2026.
Programme offering	A programme made available at a campus/mode/intake with capacity and rules.

Executive Summary

The Admission Module is the controlled entry point into the Mema College ERP. It manages the full funnel from enquiry and account creation through application, document verification, review, shortlisting, approval, offer issuance, acceptance, nomination/admission roll, fee clearance and conversion to a student record. It supports multiple

applications per person without duplicating identity records, configuration-driven programmes and intakes, auditable decisions, configurable approval paths, branded communication, dashboards, advanced analytics and standards-based APIs.

Critical design decision: Creating an account does not create a student. A student number and SIS record are generated only after an approved offer is accepted and configured enrolment gates are satisfied. Applicant numbers and QR verification tokens are separate identifiers.

Objectives

- Provide a mobile-first, accessible, secure and low-friction application journey.
- Eliminate hard-coded programmes, deadlines, counties, fees, document lists and approval chains.
- Create a single auditable source of truth for every decision, status change and communication.
- Automate admission letters, applicant numbers, QR verification, roll generation and SIS conversion.
- Provide real-time operational dashboards and governed analytics without exposing sensitive personal data.
- Support future multi-campus, multi-brand and multi-tenant evolution while launching for one institution.

Scope

In scope	Out of initial release / separately integrated
Lead capture; self-service registration; application; document upload; payments reference; review; shortlisting; approvals; offer; acceptance; admission roll; SIS conversion; communications; reporting; APIs; audit.	Full academic registration, teaching/LMS delivery, examinations, payroll, procurement and alumni operations.
Programme/intake catalog synchronisation and public website widgets.	Replacing the finance gateway, identity provider, SMS/email provider, document-signing platform or national verification services.
Applicant support, duplicate detection, deferrals, withdrawals, appeals and controlled bulk operations.	Automatic immigration or professional-body decisions without human approval.

1. Stakeholders, Actors and Roles

Actor / role	Primary responsibilities
Prospect	Search programmes, create account, consent to contact, start application.
Applicant	Maintain profile, complete applications, upload documents, pay fees, submit, track status, respond to requests, accept/decline offer.
Admissions Clerk	Validate completeness, manage enquiries, request corrections, verify non-academic data.
Admissions Officer	Review eligibility, score and recommend; manage assigned queue.
Department Reviewer	Perform academic assessment and programme-level recommendation.
Head of Department / Dean	Approve departmental recommendation where configured.
Registrar Admissions	Own admission rules, final institutional decision, admission roll and controlled overrides.
Finance Officer	Confirm application/deposit payments, reconcile exceptions and refunds.
Disability / Student Support Officer	Review disclosed support needs under restricted access.
Document Verifier	Verify identity and qualification documents; record source and outcome.

Actor / role	Primary responsibilities
Data Analyst / Management	Access aggregated dashboards and governed extracts.
System Administrator	Manage configuration, integrations, users, roles and technical operations; cannot make admission decisions by default.
Auditor / DPO	Read-only access to audit, consent, retention, access and decision evidence.
API Client / Website	Read published programme offerings or exchange authorised application events.

Segregation of duties

- No user may both change eligibility rules and approve an application affected by those changes unless an authorised emergency override is recorded.
- System administrators do not receive applicant medical/disability details unless specifically assigned a restricted support role.
- A final reject, revoke, rescind, merge, purge or override action requires elevated permission, reason code and audit event.
- The final admission roll is generated from approved records and digitally frozen; subsequent changes create a new version.

2. End-to-End Process and Status Model

1. Prospect views published programme offerings or arrives through a tracked campaign/referral.
2. Prospect creates and verifies an account, receives applicant number and may generate a QR identity/status card.
3. Applicant completes profile and one continuous, autosaved application form with section navigation and completeness indicator.
4. Applicant selects a configured programme offering, uploads required evidence and accepts the required declarations.
5. Before submission, the system presents a mandatory KES 1,000 application-fee step. The applicant selects an approved channel, completes payment and receives a reference and receipt.
6. The Submit Application button is enabled only after the gateway or authorised Finance Officer confirms the full KES 1,000 payment. The system then freezes the submitted version and routes it to review.
7. Admissions and departmental reviewers verify documents, assess eligibility, request information or make recommendations.
8. Configured approvers shortlist and approve/reject/waitlist/conditionally admit the applicant.
9. System generates a cohort-specific admission offer and secure QR verification page; applicant accepts or declines.
10. After acceptance conditions, deposit and registration gates are met, Registrar includes the candidate in the approved admission/nomination roll.
11. An idempotent conversion service creates the SIS student/person/programme-admission records and returns the student number.

Application state machine

Code	Status	Allowed next states	Meaning / gate
LEAD	Interested	ACCOUNT_CREATED, DISQUALIFIED	Optional marketing lead; no application yet.
ACCOUNT_CREATED	Account created	EMAIL_VERIFIED, LOCKED	Unique email; verification pending.
EMAIL_VERIFIED	Verified	DRAFT	Identity/contact channel verified.
DRAFT	Draft	SUBMITTED, WITHDRAWN	Editable and autosaved.
SUBMITTED	Submitted	VALIDATION_FAILED, UNDER_REVIEW, WITHDRAWN	Version frozen; acknowledgement issued.

Code	Status	Allowed next states	Meaning / gate
VALIDATION_FAILED	Correction required	DRAFT, WITHDRAWN	Applicant must correct returned sections.
UNDER_REVIEW	Under review	INFO_REQUESTED, VERIFIED, REJECTED	Assigned operational/academic checks.
INFO_REQUESTED	More information required	UNDER_REVIEW, WITHDRAWN	Clock paused until response or deadline.
VERIFIED	Verified	SHORTLISTED, REJECTED, WAITLISTED	Evidence and minimum rules passed.
SHORTLISTED	Shortlisted	APPROVAL_PENDING, REJECTED, WAITLISTED	Positive recommendation, not an offer.
APPROVAL_PENDING	Approval pending	ADMITTED_CONDITIONAL, ADMITTED, REJECTED	Configured approval chain active.
ADMITTED_CONDITIONAL	Conditional offer	ADMITTED, EXPIRED, REVOKED	Outstanding conditions are visible.
ADMITTED	Offer issued	ACCEPTED, DECLINED, EXPIRED, DEFER_REQUESTED, REVOKED	Signed admission letter available.
WAITLISTED	Waitlisted	APPROVAL_PENDING, REJECTED, WITHDRAWN	Capacity-managed queue.
ACCEPTED	Offer accepted	READY_TO_ENROL, DEFER_REQUESTED, WITHDRAWN	Acceptance declaration recorded.
READY_TO_ENROL	Cleared for enrolment	ENROLLED	All configured gates passed.
ENROLLED	Converted to student	—	SIS record and student number created.
REJECTED	Rejected	APPEALED, CLOSED	Reason and appeal information communicated.
DEFER_REQUESTED	Deferral requested	DEFERRED, ADMITTED, REJECTED	Decision required.
DEFERRED	Deferred	DRAFT/ADMITTED in new intake	New linked application/offer version.
WITHDRAWN	Withdrawn	CLOSED	Applicant-initiated; immutable history retained.
REVOKED	Offer revoked	APPEALED, CLOSED	Exceptional controlled action.
CLOSED	Closed	—	Terminal administrative closure.

3. Applicant Portal Functional Requirements

3.1 Account creation and authentication

ID	Requirement	Acceptance rule
FR-REG-001	Apply Now displays email, password, confirm password, programme search and minimum identity/contact fields.	Required fields are visibly marked; submit remains disabled until client-side minimum validation passes, with server-side validation authoritative.
FR-REG-002	Email is unique case-insensitively and verified by expiring one-time link or OTP.	No admissions access until verified; resend is rate-limited.
FR-REG-003	Passwords follow configurable length/breach/complexity policy and are stored using a current adaptive hash.	Never email or log a password; confirmation must match.
FR-REG-004	Applicant receives immutable Applicant Number after successful account creation.	Format is configurable, e.g., MC/APP/2026/000 001; generation is atomic and non-reusable.
FR-REG-005	A cryptographically random QR token resolves to	QR must not encode PII; token can be

ID	Requirement	Acceptance rule
	a minimal verification page.	rotated/revoked and access is logged.
FR-REG-006	Support login, logout, forgot/reset password, session/device revocation, optional MFA and lockout.	Reset token is single-use and expires; suspicious attempts are throttled.
FR-REG-007	Detect likely duplicate person records using email, phone, identity number and fuzzy name/DOB matching.	Do not block legitimate twins/shared contacts automatically; route possible matches for review.

3.2 Continuous application form

The applicant experiences one continuous form, visually divided into collapsible sections. Autosave writes validated partial drafts. The form supports keyboard navigation, mobile screens, back/next section controls, a sticky progress indicator, save-and-exit, validation summary, mandatory KES 1,000 payment and final review before submission.

Field / group	Canonical field	Rule	Design detail
Programme selection	programme_offering_id	Required; searchable advanced selector	Read from active ERP offerings/API cache; filter by faculty, level, mode, campus, intake, duration, study area and keyword. Show deadline, capacity status, requirements and fees.
Email	email	Required; verified; unique	Normalise lowercase; retain original display value.
Password / confirmation	password	Required at registration only	Do not persist confirmation; password never appears in reports.
Gender	gender_code	Required	Male, Female, Prefer not to say; values configurable and respectful of legal requirements.
First name	first_name	Required	Unicode letters, apostrophe and hyphen allowed; trim whitespace.
Middle name	middle_name	Optional	Do not force placeholder characters.
Last name	last_name	Required	Same name validation; support single-name exception via configured process.
Birth certificate / passport	identity_document	Required based on nationality/age/rule	Store type, number, issuing country and expiry where applicable; encrypt and mask.
Date of birth	date_of_birth	Required	Day/month/year dropdowns with search plus calendar; reject future date and apply programme age rules.
Country of residence	residence_country_code	Required; Kenya default	ISO-based searchable list; default is a convenience, not a forced value.
Phone	phone_e164	Required	Country code selector (+254 default) plus national number; normalise to E.164 and verify where configured.
Disability / additional learning needs	has_support_need	Required Yes/No	If Yes, show optional restricted details and reasonable-adjustment contact consent; never use for adverse scoring.
Place of birth	place_of_birth	Required where policy requires	Free text plus optional country/county structured fields.
Referral code	referral_code	Optional	Validate against active campaign/agent/scholarship codes; prevent applicant from editing after submission.
How did you hear about us?	source_channel_code	Required/configurable	Website; Search engine; Social media; Facebook; Instagram; TikTok; X/Twitter; LinkedIn; YouTube; Radio;

Field / group	Canonical field	Rule	Design detail
			TV; Newspaper; Education fair; School visit; Friend/family; Alumni; Employer; Agent; Email/SMS; Outdoor advert; Other.
County	county_code	Conditional	All 47 Kenyan counties plus Other/Not applicable for non-Kenyan residence; derive neither nationality nor citizenship from county.
Nationality	nationality_country_code	Required	Use searchable country list. "Kenyan/international" is a reporting classification derived from nationality, not the stored nationality value.
Acknowledgement	account_disclaimer_accepted	Required checkbox	Versioned text, timestamp, IP/device context and policy version stored.
Terms/privacy/cookies	terms_accepted, privacy_acknowledged, marketing_consent	Required terms/privacy acknowledgement; marketing optional	Separate necessary processing from optional marketing. Link versioned Terms, Privacy Notice and Cookie Notice.
Contact preference	contact_preferences	Required/configurable	Email, SMS, phone, WhatsApp where legally/operationally supported; transactional messages cannot be disabled where necessary.

3.3 Mandatory application-fee payment before submission

Applicant instruction: Before submitting your application, please pay the mandatory, non-refundable KES 1,000 application fee using one of the approved payment options shown in the portal. Your application can only be submitted after the payment has been successfully confirmed.

Requirement	Required behaviour
Amount and currency	KES 1,000 exactly. The amount is effective-dated in fee setup and fixed at KES 1,000 for this release.
Position in journey	Payment appears after the form, documents and declarations are complete, but before the final Submit Application action.
Payment channels	Support approved channels such as M-Pesa STK Push/Paybill, card, bank or cashier reference through configured providers. Display only channels enabled by Finance.
Reference	Every attempt receives a unique payment request/reference linked to the applicant and application. The payer phone/account may differ from the applicant contact.
Statuses	NOT_STARTED, INITIATED, PENDING, PAID, FAILED, CANCELLED, EXPIRED, REVERSED, REFUNDED and WAIVED. WAIVED requires an authorised Finance role and reason.
Submission gate	Submit Application remains disabled unless the authoritative state is PAID for KES 1,000 or WAIVED through an approved exception. A typed transaction code alone is not confirmation.
Confirmation	Confirm through a signed gateway callback and server-side verification. Show amount, reference, channel, payment time and status; provide a downloadable receipt.
Pending payment	Allow status refresh and safe retry without duplicate charges. A delayed callback must reconcile to the original attempt.
Failure/cancellation	Keep the application as a draft, explain the failure safely and allow another attempt or channel.
Over/under-payment	Underpayment does not unlock submission. Overpayment is flagged for Finance reconciliation and does not create credit automatically.

Requirement	Required behaviour
Refund/reversal	Payment after withdrawal/rejection follows approved Finance policy. A reversal before submission relocks the Submit Application button.
Security	Never store PIN, CVV or full card data. Use provider-controlled interfaces, signed callbacks, idempotency and replay protection.
Audit/reconciliation	Record every attempt, provider event, transition, manual confirmation/waiver, receipt and daily reconciliation exception.

3.4 Remaining application sections

Section	Core data	Key behaviours
Personal and contact	Names, previous names, DOB, gender, identity, nationality, residence, postal/physical address, county, phones, emergency contact.	Conditional fields; address reuse; identity masking; duplicate detection.
Education history	Institution, country, qualification, index/certificate number, subjects/units, grades, start/end dates.	Repeatable entries; grading scheme configuration; verification status.
Programme choices	Primary and optional alternate choices, offering, campus, mode, intake, sponsorship.	Eligibility preview; capacity/deadline check; prioritised choices; one active application per offering unless override.
Employment/experience	Employer, role, dates, responsibilities, professional memberships.	Conditional by programme; attachments supported.
References	Referee name, relationship, organisation, email, phone, consent to contact.	Configurable count; secure referee invitation/link; applicant cannot see confidential response.
Documents	Identity, photo, certificates, transcripts, result slips, recommendation, portfolio and programme-specific evidence.	Type/size/malware validation; preview; replace before submission; verifier outcome after submission.
Disability/support	Support category, requested adjustment, supporting document, preferred contact.	Restricted permission domain; not shown in general reviewer exports.
Funding/fees	Sponsor type, sponsor name, scholarship request and mandatory KES 1,000 application fee.	Show confirmed payment reference and receipt; never store card, PIN or CVV data.
Declarations	Accuracy, authenticity, data processing, discipline/criminal disclosure only when lawful, terms and signature.	Versioned, required, timestamped; electronic signature evidence.
Review, pay and submit	Read-only summary, missing items, declarations, payment state and final submit.	Submit is locked until KES 1,000 is PAID or exceptionally WAIVED; issue payment and submission receipts separately.

3.5 Applicant self-service actions

- Create, read and update draft applications; delete an unsubmitted draft only when policy permits.
- Initiate or retry the KES 1,000 application-fee payment, refresh status and download the payment receipt.
- Submit only after confirmed payment, then withdraw, request deferral, accept/decline an offer, appeal a rejection, respond to information requests and download documents.
- View an application timeline using applicant-friendly statuses; internal notes and risk flags remain hidden.
- Book interview/assessment slots where enabled and receive reminders.
- Open a support ticket linked to the application without exposing sensitive reviewer notes.
- Download a PDF application copy, payment receipt, submission receipt, offer/admission letter, fee instructions and acceptance confirmation.

4. Admissions Back-Office Workspace

Capability	Required behaviour
Work queues	My assignments, unassigned, overdue, returned, high risk, document verification, departmental review, approval, waitlist and enrolment-ready queues.
Search and filters	Applicant/application number, name, masked identity, email, phone, programme, intake, status, county, nationality class, source, reviewer, score, payment, submission date and SLA.
Application viewer	Side-by-side structured data and document preview; change history; verification badges; duplicate warnings; communication timeline.
Review checklist	Configuration-driven tasks with Pass/Fail/Not applicable, evidence, comment, reviewer and timestamp.
Scoring	Rule-based and rubric scoring with weights, minimums, reviewer moderation and explainable result.
Decisions	Recommend, shortlist, waitlist, return, approve, conditional approve, reject, defer, revoke or close subject to permission and workflow.
Bulk actions	Assign, message, shortlist, generate letters, export and roll creation only after preview, validation, reason and confirmation. No silent mass status changes.
Collaboration	Internal notes, mentions, task assignment and escalation; notes classified as general, restricted or decision evidence.
SLA controls	Timers, pause reasons, due dates, escalation and workload balancing.
Merge/duplicates	Compare and merge person records without losing application/audit history; destructive merge requires maker-checker approval.
Impersonation	Disabled by default; controlled support view-as requires reason, banner, timeout and full audit.
Soft delete/archival	Business records are never hard-deleted through normal UI. Retention/purge is a separate authorised process.

CRUD and action matrix

Entity	Applicant	Admissions	Registrar	Administrator	Auditor
Own profile	C/R/U	R/U*	R/U*	R*	R
Draft application	C/R/U/D	C/R/U*	R/U*	R*	R
Submitted application	R + response	R/U workflow	R/U/decision	R*	R
Programme offering	R	R	R/approve publish	C/R/U/archive	R
Eligibility rule	—	R/test	R/approve	C/R/U draft	R
Review/score	R outcome only	C/R/U own	R/U/approve	R metadata	R
Admission letter	R/download	generate preview	issue/revoke/reissue	template setup	R
Admission roll	—	R/draft	C/freeze/version	R	R
Audit log	own activity subset	R scoped	R	R technical	R full

* Updates to applicant-supplied data after submission require a correction workflow, stated reason and preserved before/after values.

5. Rules, Workflow and Approval Engine

Rule domain	Examples
Offering eligibility	Minimum grade, subject combinations, prior qualification, age, citizenship/residency, work experience, portfolio, professional licence.
Application completeness	Required sections/documents/declarations/payment by programme and applicant category.
Routing	Faculty/department, programme, campus, applicant type, nationality, workload, conflict-of-interest and risk.
Scoring	Weighted qualification score, subject score, interview, experience, portfolio and discretionary moderated score.
Capacity	Programme quota, campus capacity, reserved categories where lawful, waitlist priority and offer expiry.
Approval	Single or parallel departmental review; HoD/Dean/Registrar thresholds; maker-checker override.
Communication	Template/event/channel/language/reminder cadence with opt-out rules.
Conversion	Accepted offer, verified identity, condition clearance, deposit/fee state, approved roll and no blocking hold.

Approval workflow

12. Admissions Officer completes operational verification and records recommendation.
13. Department Reviewer assesses academic eligibility and provides rubric score/recommendation.
14. HoD or Dean approves the departmental recommendation where configured.
15. Registrar reviews exceptions, capacity, conditions and conflicts, then records the final institutional decision.
16. For bulk admission, a maker prepares the decision batch and a different checker approves it.
17. System issues signed documents only after the final approval transaction commits successfully.

Rule versioning: Every application is evaluated against a named, effective-dated rule-set version. Publishing a new rule never silently rewrites historical decisions. Re-evaluation is explicit, authorised and audited.

6. Admission Setups and Master Data

Setup area	Configurable items
Academic structure	Schools, faculties, departments, programmes, specialisations, award levels.
Offerings	Programme + intake + campus + mode + deadline + capacity + status + fees + eligibility rule set.
Intakes/cohorts	Code, label, year, opening/closing dates, acceptance deadline, registration dates, active/published state.
Application forms	Section/field definitions, order, help text, conditional visibility, validation, requiredness and version.
Reference data	Countries, nationalities, Kenyan counties, gender values, source channels, document types, qualifications, grading schemes.
Numbering	Applicant, application, offer, roll, receipt and student number patterns with sequence controls.

Setup area	Configurable items
Documents	Required document matrix by programme/applicant class; MIME/size limits; verification methods.
Workflow	Stages, assignee rules, approval chain, quorum, delegation, SLA and escalation.
Scoring	Criteria, weights, thresholds, rounding, reviewer count and moderation.
Capacity/waitlist	Quota, reserved capacity where lawful, ranking, tie-break rules, release schedule.
Fees/payment	Mandatory application fee KES 1,000; effective date; approved channels; provider accounts; receipt template; waiver permission; refund/reversal and reconciliation rules.
Templates	Email/SMS/letter/receipt/QR verification text, language, placeholders, signatory, effective dates.
Consent/policies	Terms, privacy, cookies, acknowledgements and optional marketing consent versions.
Integration	Website cache TTL, payment gateway, email/SMS, SIS, identity verification, storage and webhook credentials.
Retention	Record category, retention period, legal hold, anonymisation/purge workflow.
Calendars	Working days, holidays and SLA calculation calendar.

Programme publication and website caching

The ERP remains the system of record for programme offerings. The public website consumes a read-only catalogue API or signed publication feed. A cache improves performance but must not become the authority. Cache keys include locale and offering version; recommended TTL is configurable (for example 5–30 minutes), with webhook-driven invalidation on publish/unpublish. If the ERP is temporarily unavailable, the website may show the last known published catalogue with a freshness timestamp, but application submission performs a live deadline/capacity/availability check.

7. Data Model and Data Governance

Entity	Purpose
person	Canonical individual identity; supports deduplication and future student/alumni linkage.
user_account	Credentials/IdP subject, verification, lock state, MFA and sessions.
contact_point	Email/phone/address with verification, preference and validity.
applicant_profile	Admission-specific demographic and source information.
programme	Stable academic programme master.
programme_offering	Intake/campus/mode-specific published opportunity.
application	Application header, applicant, offering, current state and current submitted version.
application_version	Immutable snapshot of submitted data and declaration versions.
education_history	Repeatable prior qualifications and grades.
reference_request	Referee invitation, confidential response and status.
document_asset	Object-storage key, hash, MIME, owner, classification and malware result.

Entity	Purpose
document_verification	Outcome, method, verifier, source, notes and time.
review_assignment	Stage, reviewer, due time, conflict declaration and status.
review_score	Criterion values, weight, rationale and rubric version.
decision	Recommendation/final decision, reason code, conditions and authority.
approval_step	Workflow instance step, actor, outcome and timestamp.
offer	Offer version, acceptance deadline, status and revocation state.
generated_document	Template/version, checksum, storage key, QR token and signatory.
payment_attempt	Application, expected amount KES 1,000, channel, idempotency key, payer reference and state.
payment_transaction	Immutable provider event/reference, amount, currency, timestamp, signature-verification and reconciliation state.
payment_receipt	Unique receipt number, transaction, document checksum, issue time and storage key.
admission_roll	Versioned cohort decision list and approval evidence.
student_conversion	Idempotency key, SIS identifiers, result and error state.
communication	Template, channel, recipient, delivery status and content hash.
consent_record	Purpose, policy version, action, timestamp and evidence.
status_history	Every application state transition with actor and reason.
audit_event	Tamper-evident security/business event metadata.

Data integrity rules

- Use UUIDs internally; expose human-readable numbers separately.
- Unique constraints: normalised verified email per account; applicant number; application number; offering code; roll version; SIS conversion idempotency key.
- Submitted versions, decisions, generated letters and roll versions are immutable; corrections create new linked versions.
- All timestamps stored in UTC and displayed using the institution/user timezone.
- Identity numbers, support data and document assets are encrypted and masked; access is purpose- and role-limited.
- Files are stored outside the web root with content hashes, malware scanning, signed short-lived download URLs and retention tags.

8. APIs and Integration Design

Expose versioned REST/JSON APIs under /api/v1 with OpenAPI 3.1 documentation. Use OAuth 2.1/OIDC for users and client credentials or signed service identities for systems. Every mutation accepts a correlation ID; critical create/convert/payment endpoints support idempotency keys. Pagination is cursor-based for large collections.

Method	Endpoint	Purpose
GET	/public/programme-offerings	Published searchable catalogue; ETag/cache headers; no authentication.
GET	/public/programme-offerings/{id}	Offering details, eligibility summary and active application link.
POST	/auth/register	Create account and applicant number; rate-limited.
POST	/auth/verify-email	Verify one-time token/OTP.

Method	Endpoint	Purpose
GET/PATCH	/me/profile	Read/update applicant profile with field-level validation.
POST/GET	/applications	Create/list own applications.
GET/PATCH	/applications/{id}	Read/update draft; optimistic concurrency using version/ETag.
POST	/applications/{id}/documents	Initiate upload and register metadata; malware pipeline.
POST	/applications/{id}/submit	Atomic validation, snapshot and workflow start.
POST	/applications/{id}/withdraw	Applicant withdrawal with reason.
POST	/applications/{id}/responses	Respond to information request.
GET	/applications/{id}/timeline	Applicant-safe status history.
POST	/applications/{id}/offer-response	Accept/decline with declaration and idempotency.
GET	/verification/{qr_token}	Minimal public verification; tokenised and rate-limited.
GET	/admin/applications	Permission-filtered search/work queue.
POST	/admin/applications/{id}/assignments	Assign/reassign review stage.
POST	/admin/applications/{id}/reviews	Save checklist/score/recommendation.
POST	/admin/applications/{id}/transitions	Execute authorised state transition.
POST	/admin/decision-batches	Prepare controlled bulk decisions.
POST	/admin/decision-batches/{id}/approve	Checker approves batch.
POST	/admin/offers/{id}/issue	Generate, sign, store and communicate offer.
POST	/admin/admission-rolls	Generate roll draft from query snapshot.
POST	/admin/admission-rolls/{id}/freeze	Freeze approved roll version.
POST	/admin/applications/{id}/convert	Idempotent SIS conversion.
GET	/admin/analytics/funnel	Aggregated funnel metrics with scoped filters.
POST	/webhooks/payment	Signed provider event with replay protection.
POST	/webhooks/sis-conversion-status	Signed asynchronous SIS result.
GET	/admin/exports/{job_id}	Authorised time-limited export result.

Payment API requirements

Method	Endpoint	Purpose
GET	/applications/{id}/payment	Return expected KES 1,000, channels, attempts, authoritative status and receipt link.
POST	/applications/{id}/payment-attempts	Create an idempotent KES 1,000 payment request and provider/STK initiation.
GET	/applications/{id}/payment-attempts/{attempt_id}	Refresh a pending attempt using server-to-server provider verification.
GET	/applications/{id}/payment-receipt	Download the official receipt after PAID confirmation.
POST	/applications/{id}/submit	Validate complete form and authoritative PAID/authorised WAIVED state, then atomically snapshot and start workflow.
POST	/admin/payments/{id}/waive	Finance-only exceptional waiver with reason,

Method	Endpoint	Purpose
		evidence and maker-checker rule where configured.
POST	/admin/payments/{id}/reconcile	Resolve provider/bank/cashier exceptions without accepting a user-entered code as sole proof.
POST	/webhooks/payment	Verify provider signature, event ID, amount KES 1,000, currency KES and application reference; reject replay/out-of-order changes.

Integration events

Event	Consumers	Minimum payload
applicant.created	CRM/communications	event_id, occurred_at, applicant_id, source, consent flags
application_fee.paid	submission gate, finance, analytics	application_id, payment_id, amount=1000, currency=KES, paid_at, receipt_number
application.submitted	workflow, analytics	application_id, offering_id, version, payment_id, submitted_at
application.status_changed	notifications, analytics	from, to, actor_type, reason_code
offer.issued	communications, document service	offer_id, application_id, expiry
offer.accepted	finance, SIS readiness	offer_id, acceptance_at, conditions state
application.ready_to_enrol	SIS adapter	application/person/offering IDs, roll version
student.converted	ERP modules	person_id, student_id, student_number, programme admission ID

9. Security, Privacy and Audit

Control	Requirement
Authentication	OIDC-ready authentication, verified email, optional MFA, password breach checks, secure reset, session rotation and device/session revocation.
Authorisation	Deny-by-default RBAC plus scope/attribute checks for campus, department, intake and sensitive data domain.
Application security	CSRF protection, output encoding, parameterised queries, strict CORS, secure headers, upload isolation, rate limits and bot protection.
Encryption	TLS in transit; managed encryption at rest; field-level protection for identity/support data; secrets in a vault.
Audit	Append-only event trail for read of high-risk data, create/update, export, decision, override, impersonation, merge and purge. Capture actor, time, target, before/after hash, reason and correlation ID.
Privacy	Purpose limitation, minimisation, versioned notices/consent, subject access/correction workflows, retention, legal holds and controlled anonymisation.
Export protection	Permission, purpose, filter summary, watermark/classification, row limit, background job, expiry and audit. Sensitive columns excluded by default.
Monitoring	Alert on account enumeration, credential stuffing, high-volume exports, mass decisions, unusual document downloads, permission changes and webhook failures.

Control	Requirement
Resilience	Encrypted backups, tested restore, multi-zone deployment where available, queued communication and idempotent integration retries.
QR security	Opaque random token only; minimal verification result; revoke/rotate; rate-limit; never embed identity number, DOB or raw application data.

Recommended baseline non-functional targets

Area	Target / requirement
Availability	99.9% monthly for applicant and admissions portals excluding approved maintenance; define contractual target at deployment.
Performance	p95 catalogue/search <2s; draft save <1s server time; p95 page/API <3s under agreed peak load; submission acknowledgement <5s excluding asynchronous scans.
Scale	Design for at least 100,000 applicants/year, 5,000 concurrent public sessions and horizontal scaling; validate against actual forecasts by load test.
Recovery	Initial target RPO ≤15 minutes and RTO ≤4 hours; confirm with business impact analysis.
Accessibility	WCAG 2.2 AA; keyboard use, labels, focus order, error summary, contrast and screen-reader testing.
Compatibility	Current and previous major versions of Chrome, Edge, Firefox and Safari; responsive down to 320px.
Data quality	≥99.5% mandatory-field completeness at submission; no duplicate applicant/application numbers.
Observability	Structured logs, metrics, traces, correlation IDs, queue depth, integration health, SLA ageing and alert runbooks.
Maintainability	Configuration over code, automated tests, migration scripts, API versioning, feature flags and documented rollback.
Localisation	UTF-8, locale-aware dates/phone/currency, translatable labels/templates and timezone-safe timestamps.

10. Admission Letters, QR Verification and Printing

Cohort-specific admission letter

The letter service uses an approved effective-dated template and locks the exact rendered output. The September 2026 cohort is represented by intake configuration, not manually typed across the system. The official logo asset, contacts, signature and seal are versioned resources. Reissuing produces a new document version and records the reason; it never overwrites the previous letter.

Template placeholder	Example / source
{{college_logo}}	Approved Mema logo asset
{{college_name}}	Mema College
{{letter_reference}}	MC/ADM/SEP2026/000 123
{{issue_date}}	24 August 2026

Template placeholder	Example / source
{{applicant_name}}	Person master display name
{{applicant_number}}	MC/APP/2026/000 001
{{programme_name}}	Programme offering
{{intake_name}}	September 2026 Cohort
{{campus_mode}}	Campus and study mode
{{conditions}}	Decision conditions list
{{acceptance_deadline}}	Offer configuration
{{fee_summary}}	Approved fee schedule or secure link
{{reporting_date}}	Cohort registration schedule
{{signatory_name_title}}	Effective signatory setup
{{verification_url}}	Minimal secure QR verification page
{{document_checksum}}	Stored for authenticity validation

Letter wording control: Legal and regulatory wording, admission conditions, fee disclaimers and signatory authority must be approved by Mema College before production issuance. The system supports them; this SRDS does not invent institutional policy.

Print/export standards

- Web view: responsive, accessible and permission-scoped; printed view removes navigation and preserves headings/tables.
- PDF: tagged where feasible, embedded fonts, A4 layout, official logo/palette, page number, generation time, classification and filters.
- XLSX: real date/number types, frozen headers, filters, named sheet, data dictionary sheet and hidden export metadata; never mislabel CSV as Excel.
- CSV: UTF-8 with stable column names and ISO dates; one record type per file.
- Legacy .xls: supported only if explicitly required; warn about row limits and prefer .xlsx.
- All reports record requester, purpose, scope, filters, row count, format, checksum and expiry.

11. Analytics, Dashboards and Reports

Visualisation principle: Use 2D charts by default because they preserve accurate comparisons. 3D pie charts may be offered only as an optional presentation style for small, non-critical categories and must show the underlying values/table.

Operational dashboard

Widget / metric	Definition / use
Application funnel	Unique people/applications by stage: leads, accounts, verified, drafts, submitted, verified, shortlisted, admitted, accepted, enrolled.

Widget / metric	Definition / use
Conversion rates	Stage-to-stage conversion using a declared denominator and date basis.
Queue ageing	Open cases by current stage and age bucket; SLA breached and due soon.
Applications over time	Daily/weekly submissions and comparisons with prior intake.
Programme demand	Applications, qualified applicants, offers, acceptances, enrolments and capacity by offering.
Reviewer workload	Assigned/open/completed/overdue cases and median turnaround; not a simplistic productivity score.
Document quality	Missing/rejected/verified documents and common rejection reasons.
Payment	Expected, pending, paid, failed, waived and reconciled application/deposit payments.
Source effectiveness	Leads, submitted, admitted and enrolled by acquisition channel/campaign with cost if available.
Diversity/access	County, nationality class, gender and support-needs aggregates with privacy thresholds.
Yield forecast	Expected enrolment from offers using transparent historical rates and uncertainty band.
Integration health	Website catalogue freshness, email/SMS delivery, payment webhooks, storage scans and SIS conversion failures.

Advanced analytics

- Cohort funnel and period-over-period comparison with consistent stage definitions.
- Time-to-stage, review-cycle time, bottleneck and survival/ageing analysis.
- Programme demand-to-capacity modelling and waitlist release scenarios.
- Geographic distribution by county/country using privacy-safe aggregation.
- Campaign attribution using first-touch, last-touch and declared source; label assumptions.
- Offer yield and enrolment forecast with confidence intervals and back-testing.
- Duplicate/fraud anomaly indicators for human review; never automatic rejection.
- Equity monitoring of outcomes by lawful demographic categories; suppress small cells and investigate process bias.
- Applicant drop-off analysis by form section, device and validation error without capturing password/document content.
- What-if cohort planning using capacity, acceptance rate, expected deferral and target enrolment.

Standard report catalogue

Report	Purpose
Applicant register	All authorised applicant/application records by intake/programme/status.
Application funnel	Stage counts and conversion rates by time/cohort.
Incomplete applications	Drafts with missing sections and last activity; contact preferences applied.
Submitted applications	Submission register and acknowledgement status.
Document verification	Missing, rejected, verified and pending documents.
Eligibility and scoring	Rule outcome, criterion scores and reviewer recommendations.
Shortlist	Approved shortlist version with selection basis.
Decision report	Offers, conditional offers, waitlists and rejections with governed reason

Report	Purpose
	codes.
Admission roll / nomination roll	Frozen approved candidate list by cohort/programme; versioned and signed.
Offer acceptance	Accepted, declined, expired and pending; deadline ageing.
Enrolment conversion	Ready, converted, failed and reconciled student numbers.
Fees/reconciliation	Application/deposit payment state and exceptions.
Source/campaign	Declared and tracked acquisition performance.
County/nationality	Privacy-safe geographic profile.
SLA/workload	Stage turnaround, queue ageing, assignments and breaches.
Communication delivery	Email/SMS/WhatsApp status, bounce/failure and resend.
Audit/override	Sensitive access, exports, permission changes, decision overrides and merges.
Data-quality exceptions	Duplicates, invalid contacts, orphaned data and integration discrepancies.

Metric governance

- Each metric has an owner, formula, grain, allowed dimensions, source tables, refresh schedule and privacy rule.
- Dashboards show last refresh time, applied filters and whether counts are persons or applications.
- Exports reproduce dashboard filters and provide a machine-readable data dictionary.
- Small-cell suppression and role-based aggregation prevent re-identification.
- Historical trend uses event/status history, not only the current application status.

12. Notifications and Communications

Trigger	Audience	Channels / content
Account created / verify	Applicant	Email/OTP, applicant number, verification and security advice.
Draft inactivity	Applicant	Consent-aware reminder; programme deadline; resume link.
Submitted	Applicant + queue	Receipt, application number, timestamp and expected next step.
Information requested	Applicant	Specific items, secure response link and deadline.
Status decision	Applicant	Approved template; avoid sensitive reason in SMS.
Offer issued	Applicant	Admission letter, acceptance deadline and secure response.
Offer expiring	Applicant + officer	Reminder schedule and escalation.
Accepted / ready	Applicant + finance/registrar	Confirmation, outstanding conditions and registration instructions.
Converted	Student + integrated systems	Student number and next-step onboarding.
Integration/security failure	ICT/support	Operational alert with correlation ID, no excessive PII.

13. Error Handling and Edge Cases

- Expired programme deadline between draft and submit: preserve draft, prevent submission and offer eligible alternative/appeal path.
- Programme unpublished: prevent new selection; retain existing application and route per policy.
- Duplicate submit/double click: idempotency returns the same receipt and creates one workflow.
- Upload interrupted or infected: quarantine, explain safely and allow replacement; never expose scanner details.
- Payment webhook replay/out-of-order event: validate signature, event ID and state transition.
- Reviewer conflict or absence: declaration, recusal, reassignment/delegation and preserved SLA evidence.
- Offer capacity exhausted after approval: transactionally reserve/release seats and route exception instead of oversubscription.
- SIS unavailable: queue retry with idempotency; show “conversion pending,” never generate a second student.
- Name/identity correction after offer: controlled correction, re-verification and document reissue version.
- Applicant is a minor: show configured guardian/consent section where legally required.
- International applicant: replace Kenyan county requirement with Not applicable/Other and show passport/visa-specific documents.
- Accessibility or low bandwidth: resumable uploads, compressed pages, autosave, clear timeouts and support channel.

14. Deployment, Operations and Administration

Area	Requirement
Environments	Separate development, test/UAT, staging and production; masked/synthetic data outside production.
Release	Versioned migrations, automated regression/security tests, change approval, rollback and release notes.
Jobs	Queues for documents, email/SMS, analytics refresh, malware scans, exports and integrations; retry with dead-letter handling.
Storage	Private object storage, lifecycle policy, checksums, antivirus status and backup coverage.
Backups	Encrypted scheduled database/file backups, off-site copy, restore drills and documented recovery ownership.
Administration	Health dashboard, queue replay controls, integration status, configuration promotion, feature flags and audit search.
Support	Applicant case ID, knowledge base, ticket linkage, escalation matrix and service hours.
Data migration	Mapping, cleansing, duplicate resolution, reconciliation counts, sign-off and rollback for legacy applicants.
Training	Role-based training for admissions, reviewers, approvers, finance, analysts, support and administrators.
Go-live	Pilot intake, load/security/accessibility testing, approved templates/rules, data protection review, support roster and hypercare.

15. User Stories and Acceptance Criteria

ID	As a	I want to	So that	Acceptance criteria
US-01	Prospect	search published programmes using filters	I can identify an eligible offering before registering	Only active/published offerings appear; result shows intake, mode, campus, deadline and summary eligibility.

ID	As a	I want to	So that	Acceptance criteria
US-02	Applicant	create and verify an account	my application is secure and recoverable	Unique verified email, password policy, applicant number and acknowledgement are recorded.
US-03	Applicant	complete one continuous autosaved form	I can pause and resume without losing work	Every section shows completeness; drafts save safely; server-side validation appears by field and in summary.
US-04	Applicant	pay KES 1,000 and submit a complete application once	I receive proof of payment and submission	The gateway confirms PAID before submission; payment and submission receipts, snapshot, status and communication are created exactly once.
US-05	Admissions Officer	work from a filtered queue	I can process cases within SLA	Permission-scoped filters, saved views, assignment and ageing are available.
US-06	Document Verifier	preview and verify evidence	the decision uses authentic documents	Outcome, method, note, verifier and time are retained; file remains immutable after submission.
US-07	Department Reviewer	score with the published rubric	recommendations are consistent and explainable	System calculates weighted result, validates completeness and stores rubric version.
US-08	Registrar	approve or reject with reason and conditions	the institution retains decision authority	Permission, separation, reason, conditions, history and communication template are enforced.
US-09	Applicant	download and verify my admission letter	I can prove the offer is authentic	PDF includes correct cohort/template and QR opens minimal current verification.
US-10	Applicant	accept or decline before the deadline	my response is formally recorded	Declaration version, timestamp and single final response are stored; exception flow exists.
US-11	Registrar	freeze an admission roll	the cohort list is controlled	Roll has query snapshot, version, totals, approvals, checksum and immutable export.
US-12	Integration service	convert cleared candidates idempotently	there are no duplicate students	Repeated request returns same SIS identifiers; failures are retryable and reconciled.
US-13	Analyst	view funnel and export governed data	management can make informed decisions	Metrics show definition, date basis, filters and freshness; export is scoped and audited.
US-14	DPO/Auditor	trace access and decisions	compliance evidence is available	Audit search links actor, action, target, time, reason, correlation and before/after evidence.
US-15	Support Officer	help without viewing restricted information	privacy is preserved	Support view masks identity/support data and records every elevated access.

16. UAT Strategy

UAT validates configured business behaviour in a production-like environment using synthetic or approved masked data. A test passes only when the expected result, audit evidence and downstream effects all match. Defects are logged with severity, evidence and retest result; “actual result” remains blank until execution.

Severity	Definition	Exit treatment
Critical	Security/privacy breach, data loss, duplicate student, unusable submission or incorrect mass decision.	Zero open.
High	Major workflow, payment, letter, approval or	Zero open unless formally waived by sponsor

Severity	Definition	Exit treatment
	integration failure without safe workaround.	and DPO/security where relevant.
Medium	Business function impaired with acceptable temporary workaround.	Documented plan/owner/date.
Low	Cosmetic or minor usability issue.	May be deferred with approval.

UAT entry and exit criteria

Entry	Exit
Approved SRDS, configured test intake/programmes/rules/templates, stable build, test accounts/data, integrations or stubs, defect process and trained testers.	All planned critical/high-risk cases executed; critical/high defects closed; security/accessibility/load evidence accepted; reconciliation complete; business, ICT and DPO sign-off recorded.

17. UAT Test Cases

Test ID	Scenario	Actor	Steps / test data	Expected result
TC-REG-001	Register with valid mandatory data	Applicant	Enter valid email/password/programme/identity/contact, accept required notices, create account.	Account and applicant number created once; verification sent; QR token created; status ACCOUNT_CREATED.
TC-REG-002	Mandatory fields omitted	Applicant	Leave each required field empty and attempt submit.	Inline errors and summary shown; create disabled where appropriate; no account created.
TC-REG-003	Invalid/duplicate email	Applicant	Use malformed email then an existing verified email.	Format rejected; duplicate handled without account enumeration; recovery guidance shown.
TC-REG-004	Password policy and mismatch	Applicant	Use weak/breached password and mismatching confirmation.	Clear policy error; password never logged or emailed.
TC-REG-005	Email verification expiry/resend	Applicant	Use expired token, resend, then use newest token.	Expired rejected; resend throttled; prior token invalidated; status EMAIL_VERIFIED.
TC-REG-006	Applicant number concurrency	System	Create multiple accounts concurrently.	Unique sequential-format applicant numbers; no duplicates or gaps caused by rollback beyond documented sequence behaviour.
TC-PRG-001	Programme advanced search	Applicant	Filter by keyword, level, mode, campus and September 2026.	Only matching published offerings appear with accurate deadline/requirements.
TC-PRG-002	Catalogue cache invalidation	Administrator	Unpublish an offering and trigger publication event.	Website cache refreshes within configured target; new applications blocked; existing cases handled by rule.
TC-FRM-001	Autosave and resume	Applicant	Complete sections, close session and return.	Latest valid draft restored with progress; no duplicate repeatable records.
TC-FRM-002	DOB controls	Applicant	Use dropdown/calendar; try future date and invalid date.	Valid date stored in ISO form; invalid/future blocked; age rule explained.
TC-FRM-003	International applicant conditions	Applicant	Select non-Kenyan nationality/residence and passport.	County allows Not applicable/Other; correct phone/document fields appear; no forced Kenyan ID.
TC-FRM-004	Kenyan county list	Applicant	Open county selector and search.	All 47 counties available once, plus Other/Not applicable per

Test ID	Scenario	Actor	Steps / test data	Expected result
				rule.
TC-FRM-005	Phone normalisation	Applicant	Select +254 and enter common national formats.	Stored E.164 value is correct; invalid length rejected; verification works where enabled.
TC-FRM-006	Support needs privacy	Applicant/Officer	Select Yes and enter support details; view as ordinary reviewer.	Restricted details saved but masked/hidden from unauthorised reviewer and exports.
TC-FRM-007	Source and referral	Applicant	Select Facebook and enter valid/invalid referral codes.	Source saved; valid campaign linked; invalid code explained; optional code may be removed before submission.
TC-FRM-008	Declarations and consent	Applicant	Attempt submission without required acknowledgement; toggle marketing.	Required declaration blocks submit; marketing remains optional; versions/timestamps recorded separately.
TC-DOC-001	Valid document upload	Applicant	Upload allowed PDF/JPG within limit.	File scanned, hashed, privately stored and previewable by authorised users.
TC-DOC-002	Unsafe/invalid upload	Applicant	Upload executable, oversize file and malware test file.	Rejected/quarantined safely; no execution; event logged; applicant receives safe guidance.
TC-DOC-003	Replace before/after submission	Applicant	Replace a draft file, submit, then attempt replacement.	Draft replacement allowed; submitted evidence immutable; correction request creates controlled new version.
TC-PAY-001	Successful KES 1,000 payment before submission	Applicant	Complete the form, select M-Pesa/approved channel and pay KES 1,000.	Signed provider confirmation sets PAID; receipt generated; Submit Application becomes enabled.
TC-PAY-002	Submission attempted without payment	Applicant	Complete all form fields but leave fee NOT_STARTED/PENDING and attempt submit through UI and API.	Submit remains disabled/HTTP validation rejected; application stays DRAFT; no workflow starts.
TC-PAY-003	Underpayment and wrong currency	Applicant/System	Provider reports less than KES 1,000 or a non-KES currency.	Payment not accepted as PAID; submission locked; Finance exception created.
TC-PAY-004	Duplicate callback and repeated payment request	System	Replay the same provider event and retry with the same idempotency key.	One transaction and receipt only; no double charge/credit; replay recorded.
TC-PAY-005	Pending, failed and cancelled attempts	Applicant	Simulate timeout, failed STK/card and cancellation, then retry.	Application stays DRAFT; safe status shown; retry succeeds without losing data or duplicating attempts.
TC-PAY-006	Payment reversal before submission	Finance/System	Confirm PAID, then receive valid reversal before submission.	State changes to REVERSED; receipt marked accordingly; Submit Application relocks; audit/notification created.
TC-PAY-007	Authorised waiver exception	Finance Officer	Attempt waiver without permission, then approve with authorised role/reason.	Unauthorised action denied; authorised WAIVED unlocks submit and is fully audited.
TC-PAY-008	Payment receipt and reconciliation	Applicant/Finance	Download receipt and reconcile provider settlement file.	Receipt shows applicant/application, KES 1,000, reference, channel and time; totals reconcile and exceptions are listed.
TC-SUB-001	Complete submission	Applicant	Complete all configured sections, documents, fee and declarations; submit.	Immutable version and receipt created; workflow starts; SUBMITTED status; confirmation delivered.
TC-SUB-002	Double submit / retry	Applicant	Double-click and retry same idempotency key.	One submission/version/workflow/receipt only.
TC-SUB-003	Deadline passes before submit	Applicant	Submit immediately after	Submission blocked using server

Test ID	Scenario	Actor	Steps / test data	Expected result
			configured closing time.	time; draft retained; exception/alternative information shown.
TC-REV-001	Queue assignment and SLA	Admissions Lead	Assign by department and inspect due dates.	Only eligible reviewer receives task; SLA uses configured working calendar; audit created.
TC-REV-002	Conflict of interest	Reviewer	Declare conflict on assigned case.	Reviewer cannot decide; case reassigned; conflict stored with restricted visibility.
TC-REV-003	Checklist and score	Reviewer	Complete rubric with boundary values and recommendation.	Weights/threshold/rounding correct; incomplete rubric cannot submit; version retained.
TC-REV-004	Information request cycle	Officer/Applicant	Request specific correction; applicant responds.	Status and SLA pause/resume correctly; response linked; history visible appropriately.
TC-APR-001	Approval chain	Officer/HoD/Registrar	Recommend, approve departmentally, then final approve.	Only configured sequence/roles allowed; final decision creates offer event.
TC-APR-002	Unauthorised decision	Administrator	Attempt final approval without Registrar permission.	403/denied UI action; no state change; security event logged.
TC-APR-003	Maker-checker bulk decision	Admissions/Registrar	Prepare batch then attempt self-approval and separate approval.	Self-approval blocked; checker sees counts/exceptions; approved batch changes exactly scoped records.
TC-APR-004	Conditional offer clearance	Registrar/Applicant	Issue conditions, receive evidence, clear all.	Conditions visible/versioned; ADMITTED only after authorised clearance.
TC-OFR-001	Generate September 2026 letter	Registrar	Issue offer from approved September 2026 application.	Correct logo/config, cohort, programme, reference, signatory, conditions and checksum; stored immutable PDF.
TC-OFR-002	QR verification and revocation	Public/Registrar	Scan QR before and after offer revocation.	Minimal valid status initially; revoked status after action; no sensitive PII; scans rate-limited/logged.
TC-OFR-003	Accept/decline/expiry	Applicant	Accept before deadline; repeat response; test expired offer.	One valid response; repeat idempotent; expired offer needs authorised extension; declaration evidence retained.
TC-ROL-001	Generate and freeze roll	Registrar	Create roll from approved query, review totals and freeze.	Only eligible records included; query snapshot/version/checksum/app rovals retained; frozen roll immutable.
TC-SIS-001	Successful SIS conversion	Registrar/System	Convert READY_TO_ENROL candidate.	Person/student/programme records created; student number returned; status ENROLLED.
TC-SIS-002	Idempotent retry and outage	System	Repeat conversion and simulate SIS timeout.	Repeat returns same identifiers; outage queues retry; no duplicate student; reconciliation alert created.
TC-RPT-001	Dashboard metric reconciliation	Analyst	Compare funnel widgets with authorised record-level extract.	Counts reconcile by documented person/application grain; filters and refresh time shown.
TC-RPT-002	Exports	Authorised user	Export same report to web, XLSX, CSV and PDF.	Values/filters match; .xlsx types/headers valid; UTF-8 CSV; branded legible PDF; audit and expiry recorded.
TC-RPT-003	Small-cell/privacy suppression	Analyst	Filter sensitive demographic dimensions to small group.	Values suppressed/aggregated per threshold; export cannot bypass suppression.
TC-SEC-001	RBAC and scoped access	Users	Attempt cross-department/campus access and direct object URL.	Denied consistently in UI/API/file layer; event logged.

Test ID	Scenario	Actor	Steps / test data	Expected result
TC-SEC-002	Rate limit and enumeration	External	Rapidly test login/register/reset/QR endpoints.	Throttled with generic responses; legitimate recovery remains usable; alert threshold works.
TC-AUD-001	Audit completeness	Auditor	Trace profile correction, review, override, export and letter issue.	Actor, time, target, action, reason, correlation and evidence hashes are complete and immutable.
TC-A11Y-001	Keyboard/screen reader	Applicant	Complete critical journey without mouse and with screen reader.	Logical focus, labels, error summary, announcements, contrast and no keyboard trap; WCAG 2.2 AA evidence.
TC-PERF-001	Peak load	Performance team	Execute agreed catalogue, draft-save, upload and submission mix at peak concurrency.	Targets met without data loss/duplicates; queues recover; resource/error metrics within thresholds.
TC-DR-001	Backup restore	Operations	Restore selected backup to isolated environment and reconcile.	RPO/RTO target met; records/files/checksums and key workflows verified.
TC-COM-001	Notification delivery/failure	System	Trigger submission and offer messages; simulate bounce/provider outage.	Correct template/recipient; no sensitive SMS content; retry/failure status and operator alert recorded.
TC-DUP-001	Possible duplicate and merge	Admissions/Registrar	Create near-match person; review and merge with checker.	Warning does not auto-reject; merge preserves all links/history and requires authorised second approval.
TC-RET-001	Retention/legal hold	DPO	Run eligible purge with and without legal hold.	Held record excluded; authorised purge/anonymisation produces audit and reconciliation evidence.

UAT execution record

Test ID	Tester	Date	Actual result / evidence	Status	Defect ID
TC-REG-001				Not run	
TC-REG-002				Not run	
TC-REG-003				Not run	
TC-REG-004				Not run	
TC-REG-005				Not run	
TC-REG-006				Not run	
TC-PRG-001				Not run	
TC-PRG-002				Not run	
TC-FRM-001				Not run	
TC-FRM-002				Not run	
TC-FRM-003				Not run	
TC-FRM-004				Not run	
TC-FRM-005				Not run	

Test ID	Tester	Date	Actual result / evidence	Status	Defect ID
TC-FRM-006				Not run	
TC-FRM-007				Not run	
TC-FRM-008				Not run	
TC-DOC-001				Not run	
TC-DOC-002				Not run	
TC-DOC-003				Not run	
TC-PAY-001				Not run	
TC-PAY-002				Not run	
TC-PAY-003				Not run	
TC-PAY-004				Not run	
TC-PAY-005				Not run	
TC-PAY-006				Not run	
TC-PAY-007				Not run	
TC-PAY-008				Not run	
TC-SUB-001				Not run	
TC-SUB-002				Not run	
TC-SUB-003				Not run	
TC-REV-001				Not run	
TC-REV-002				Not run	
TC-REV-003				Not run	
TC-REV-004				Not run	
TC-APR-001				Not run	
TC-APR-002				Not run	
TC-APR-003				Not run	
TC-APR-004				Not run	
TC-OFR-001				Not run	
TC-OFR-002				Not run	
TC-OFR-003				Not run	
TC-ROL-001				Not run	
TC-SIS-001				Not run	
TC-SIS-002				Not run	
TC-RPT-001				Not run	

Test ID	Tester	Date	Actual result / evidence	Status	Defect ID
TC-RPT-002				Not run	
TC-RPT-003				Not run	
TC-SEC-001				Not run	
TC-SEC-002				Not run	
TC-AUD-001				Not run	
TC-A11Y-001				Not run	
TC-PERF-001				Not run	
TC-DR-001				Not run	
TC-COM-001				Not run	
TC-DUP-001				Not run	
TC-RET-001				Not run	

18. Requirements Traceability Matrix

Capability	Requirement reference	User stories	UAT coverage
Account and identity	FR-REG-001–007	US-01–03	TC-REG-001–006, TC-SEC-002
Programme catalogue/cache	Section 6	US-01	TC-PRG-001–002
Continuous application	Sections 3.2–3.5	US-03	TC-FRM-001–008
Application payment	Section 3.3 and Section 8	US-04	TC-PAY-001–008
Documents/submission	Sections 3.4–3.5	US-04, US-06	TC-DOC-001–003, TC-SUB-001–003
Review/approvals	Sections 4–5	US-05–08	TC-REV-001–004, TC-APR-001–004
Letters/QR/offers	Section 10	US-09–10	TC-OFR-001–003
Roll/SIS conversion	Sections 2, 8	US-11–12	TC-ROL-001, TC-SIS-001–002
Analytics/reports	Section 11	US-13	TC-RPT-001–003
Security/privacy/audit	Sections 7, 9	US-14–15	TC-SEC-001–002, TC-AUD-001, TC-RET-001
NFR/operations	Sections 9, 14	All	TC-A11Y-001, TC-PERF-001, TC-DR-001, TC-COM-001

19. Implementation Priorities and Release Plan

Release	Scope	Exit evidence
MVP / R1	Setups; catalogue; registration/verification; continuous form; uploads; mandatory KES 1,000 payment gate and receipt; submission; queues; basic review/approval; offer letter/QR; acceptance; core reports/audit.	End-to-end happy path, payment reconciliation, security/accessibility baseline, reconciled UAT and pilot intake.
R2	Advanced scoring, waitlist/capacity, enhanced	Maker-checker, idempotency, integration and DR

Release	Scope	Exit evidence
	payment exception automation, admission roll, SIS conversion, bulk decisions, enhanced dashboards and communication automation.	tests.
R3	Referee portal, interview scheduling, deferrals/appeals, campaign attribution, forecasting, anomaly detection and richer self-service.	Model governance, fairness/privacy review and operational adoption metrics.

Recommended build order

18. Confirm policies, lifecycle, programme/intake masters, numbering and role model.
19. Agree API contracts and data model; create event/audit foundations.
20. Build public catalogue, identity and continuous draft application.
21. Add document service, validation, submission snapshot and workflow queues.
22. Implement review, approval, letter/QR and applicant decision response.
23. Implement admission roll, SIS conversion, reporting/export and reconciliations.
24. Complete security, privacy, accessibility, performance, DR, UAT and controlled pilot.

20. Open Decisions and Assumptions

Item	Current assumption / decision required	Owner
Official identity	Mema College legal name, logo, addresses and authorised signatories are placeholders.	Executive/Registrar
Brand palette	Professional teal/gold used provisionally; approve official RGB/CMYK/font rules.	Communications
Application fee	Confirmed as mandatory KES 1,000 before submission. Finance must approve channels, settlement account, waiver authority, refund/reversal rules and non-refundable wording.	Finance
Approval chain	Officer → Department → HoD/Dean → Registrar is configurable, not fixed.	Registrar
Student-number timing	Generated only after admission roll/acceptance/enrolment gates, not account creation.	Registrar/ICT
Identity verification	Document workflow included; external authoritative services require contracts/API confirmation.	Registrar/DPO
Data retention	No period invented; approve schedule under applicable Kenyan law and institutional policy.	DPO/Legal
Infrastructure SLOs	Initial targets are baselines and require load/business impact validation.	ICT
Communication channels	Email/SMS supported; WhatsApp depends on approved provider and consent model.	ICT/Communications
Legacy .xls	Prefer .xlsx; retain .xls only if a dependent process proves necessary.	Business Owner

21. Final Acceptance and Sign-off

By signing, stakeholders confirm that this SRDS accurately describes the intended Admission Module baseline, that open decisions have owners, and that changes after approval will follow controlled change management. Approval of this

document does not substitute for legal, data-protection, cybersecurity, accessibility, performance or operational readiness reviews.

Role	Name	Decision / conditions	Signature	Date
Executive Sponsor				
Registrar / Product Owner				
ICT Lead				
Finance Lead				
Data Protection Officer				
UAT Lead				